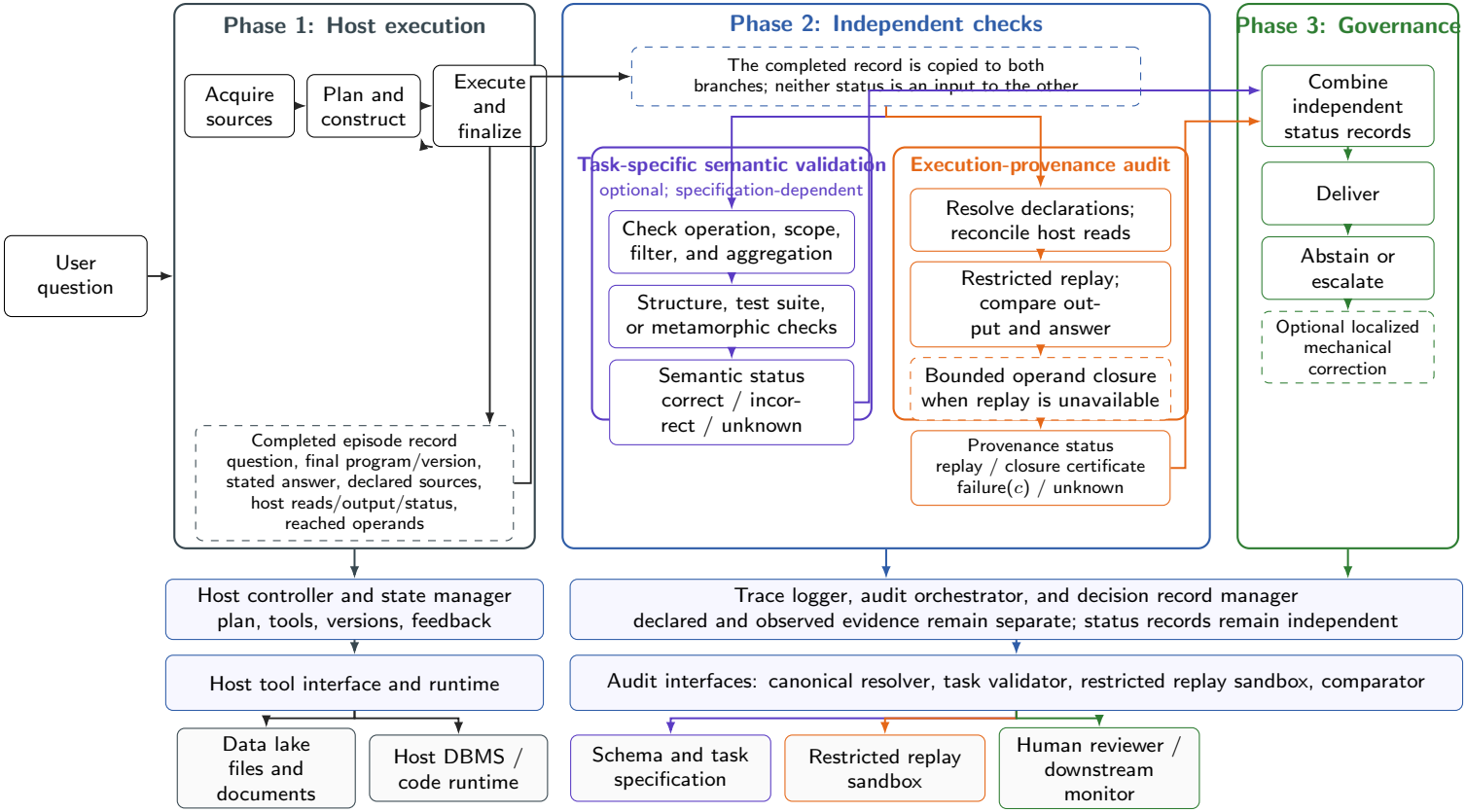
Phase 1: Host execution
Acquire sources
Plan and construct
Execute and finalize
User question
Completed episode record
question, final program/version,
stated answer, declared sources,
host reads/output/status,
reached operands
Phase 2: Independent checks
The completed record is copied to both
branches; neither status is an input to the other
Task-specific semantic validation
optional; specification-dependent
Check operation, scope, filter, and aggregation
Structure, test suite, or metamorphic checks
Semantic status correct / incorrect / unknown
Execution-provenance audit
Resolve declarations; reconcile host reads
Restricted replay; compare output and answer
Bounded operand closure when replay is unavailable
Provenance status
replay / closure certificate
failure(c) / unknown
Phase 3: Governance
Combine independent status records
Deliver
Abstain or escalate
Optional localized mechanical correction
Host controller and state manager
plan, tools, versions, feedback
Host tool interface and runtime
Data lake files and documents
Host DBMS / code runtime
Trace logger, audit orchestrator, and decision record manager
declared and observed evidence remain separate; status records remain independent
Audit interfaces: canonical resolver, task validator, restricted replay sandbox, comparator
Schema and task specification
Restricted replay sandbox
Human reviewer / downstream monitor